

## Unit 4 Guided Notes — Geometry: Angles & Triangles

Name the relationship, write the equation, solve for  $x$ . Angles always add up to something.

Standard: NC.8.G.5

 [Blank Worksheet \(PDF\)](#)

 [Answer Key \(PDF\)](#)

 [Official PDF Packet](#)

 [EC/Modified Notes \(PDF\)](#)

This packet has three parts: **Part 1 — Guided Notes** (fill in the blanks as you learn), **Part 2 — Practice Problems**, and the **Answer Key** built right in. Name the relationship, write the equation, solve for  $x$  — angles always add up to something!

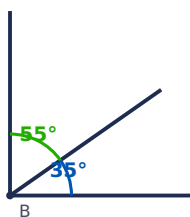
### 1 Angle Relationships

Fill the definitions and rules on your printed copy, then click to check.

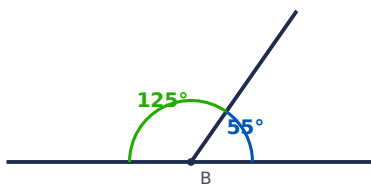
| Relationship    | Definition                           | Sum / Rule          |
|-----------------|--------------------------------------|---------------------|
| Complementary   | Two angles that add to $90^\circ$    | $a + b = 90^\circ$  |
| Supplementary   | Two angles that add to $180^\circ$   | $a + b = 180^\circ$ |
| Vertical Angles | Opposite angles when two lines cross | Always EQUAL        |
| Linear Pair     | Adjacent angles on a straight line   | Add to $180^\circ$  |

Each angle is named with three letters — the vertex is the MIDDLE letter (e.g.  $\angle ABD$  has vertex B). Arcs show which angle each measure belongs to.

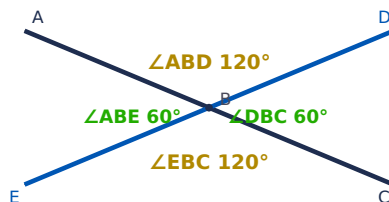
COMPLEMENTARY =  $90^\circ$



SUPPLEMENTARY =  $180^\circ$

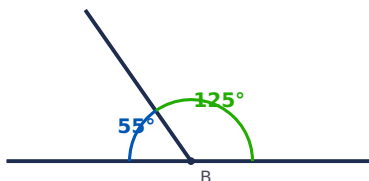


### VERTICAL ANGLES



Opposite angles are EQUAL:  $\angle ABD = \angle EBC$ ,  $\angle DBC = \angle ABE$

### LINEAR PAIR = $180^\circ$



### Quick Check

**MC** Two angles form a linear pair. One measures  $115^\circ$ . What is the other?

✓ A linear pair adds to  $180^\circ$ .  $180^\circ - 115^\circ = 65^\circ$ .

### Match the Relationship to its Rule

**MATCH** Click a relationship, then its rule.

#### Relationship

Complementary

Supplementary

Vertical Angles

Linear Pair

#### Rule

Always EQUAL

Add to  $90^\circ$

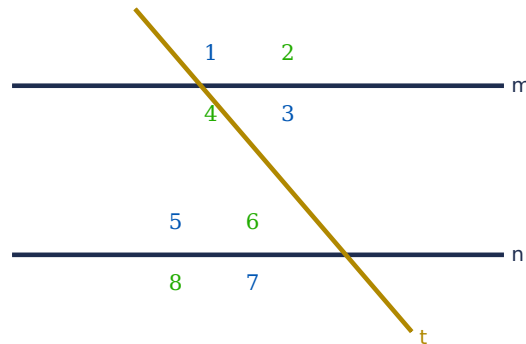
Adjacent angles on a line  $\rightarrow 180^\circ$

Add to  $180^\circ$

0 of 4 matched

✦ **REMEMBER:** Corresponding, Alternate Interior, Alternate Exterior, and Vertical angles are **CONGRUENT** (equal). Same-Side Interior and Same-Side Exterior angles are **SUPPLEMENTARY** (add to  $180^\circ$ ).

### PARALLEL LINES & TRANSVERSAL

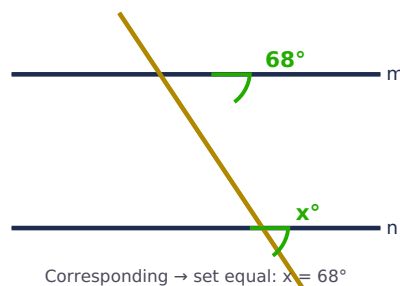


Two parallel lines ( $m \parallel n$ ) cut by transversal  $t$  form 8 angles. Corresponding: (1,5)(2,6)(3,7)(4,8) · Alt. Interior: (3,6)(4,5) · Alt. Exterior: (1,8)(2,7) · Same-Side Interior: (3,5)(4,6)

Fill the Relationship and Rule columns on your printed copy, then click to check.

| Angle Pair Type    | Relationship         | Rule to Use                            | Example                               |
|--------------------|----------------------|----------------------------------------|---------------------------------------|
| Corresponding      | <b>Congruent</b>     | <b>Set equal</b>                       | $68^\circ \rightarrow x = 68^\circ$   |
| Alternate Interior | <b>Congruent</b>     | <b>Set equal</b>                       | $101^\circ \rightarrow x = 101^\circ$ |
| Alternate Exterior | <b>Congruent</b>     | <b>Set equal</b>                       | $142^\circ \rightarrow x = 142^\circ$ |
| Vertical           | <b>Congruent</b>     | <b>Set equal</b>                       | $120^\circ \rightarrow x = 120^\circ$ |
| Same-Side Interior | <b>Supplementary</b> | <b><math>180 - \text{given}</math></b> | $134^\circ \rightarrow x = 46^\circ$  |
| Same-Side Exterior | <b>Supplementary</b> | <b><math>180 - \text{given}</math></b> | $77^\circ \rightarrow x = 103^\circ$  |

### CORRESPONDING ANGLES ARE EQUAL



**Worked Example** —  $\angle 1 = 3x + 15$  and  $\angle 2 = 5x - 7$  (alternate interior angles)

Alternate interior angles are **EQUAL**:  $3x + 15 = 5x - 7$

Subtract  $3x$ :  $15 = 2x - 7$

Add  $7$ :  $22 = 2x$

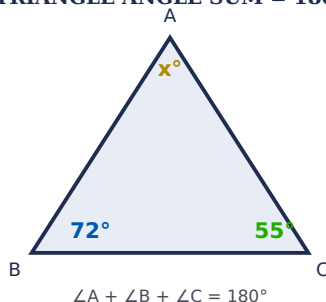
✓ Divide by  $2$ :  $x = 11 \rightarrow \text{angle} = 3(11) + 15 = 48^\circ$

### 3 Triangle Theorems

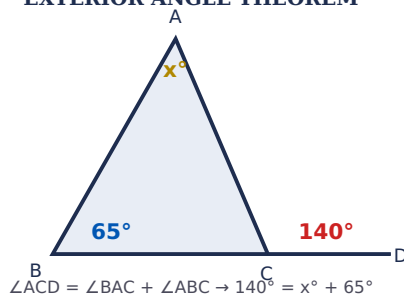
🔑 **TRIANGLE ANGLE SUM:** The three interior angles of a triangle always add up to  $180^\circ$ .

🔑 **EXTERIOR ANGLE THEOREM:** An exterior angle equals the SUM of the two non-adjacent (remote) interior angles.

TRIANGLE ANGLE SUM =  $180^\circ$



EXTERIOR ANGLE THEOREM



**Problem 1 — Triangle Angle Sum:**  $x + 72 + 55 = 180$

**Solve:**  $x + 127 = 180 \rightarrow x = 53^\circ$

**Problem 2 — Exterior Angle Theorem:**  $x + 65 = 140$

**Solve:**  $x + 65 = 140 \rightarrow x = 75^\circ$

**ERROR ANALYSIS** Jordan's Exterior Angle Mistake — A triangle has interior angles  $35^\circ$  and  $70^\circ$ . Exterior angle =  $x$ . Jordan wrote  $x = 70 - 35 = 35^\circ$ .

Jordan subtracted instead of adding. The exterior angle = SUM of the remote interior angles:  $x = 35 + 70 = 105^\circ$ .

➡ Go to Unit 4 Practice →

✦ **Identify the relationship, write the equation, and solve for  $x$ .** Type just the number, e.g. 55.

### A. Angle Relationships — Solve for $x$

**1) Supplementary Angles, given  $125^\circ$ .  $x = ?$**

$x =$

**2) Complementary Angles, given  $47^\circ$ .  $x = ?$**

$x =$

**3) Vertical Angles, given  $72^\circ$ .  $x = ?$**

$x =$

**4) Linear Pair, given  $138^\circ$ .  $x = ?$**

$x =$

**5) Complementary Angles, given  $35^\circ$ .  $x = ?$**

$x =$

**6) Supplementary Angles, given  $110^\circ$ .  $x = ?$**

$x =$

**7) Vertical Angles, given  $63^\circ$ .  $x = ?$**

$x =$

**8) Linear Pair, given  $157^\circ$ .  $x = ?$**

$x =$

**9) Complementary Angles, given  $52^\circ$ .  $x = ?$**

$x =$

$x =$

**10) Supplementary Angles, given  $95^\circ$ .  $x = ?$**

$x =$

## B. Parallel Lines & Transversals

**1) Corresponding, given  $65^\circ$ .  $x = ?$**

$x =$

**2) Alternate Interior, given  $110^\circ$ .  $x = ?$**

$x =$

**3) Alternate Exterior, given  $75^\circ$ .  $x = ?$**

$x =$

**4) Corresponding, given  $48^\circ$ .  $x = ?$**

$x =$

**5) Alternate Interior, given  $130^\circ$ .  $x = ?$**

$x =$

**6) Co-Interior, given  $55^\circ$ .  $x = ?$**

$x =$

**7) Alternate Exterior, given  $92^\circ$ .  $x = ?$**

$x =$

**8) Co-Interior, given  $140^\circ$ .  $x = ?$**

$x =$

## C. Triangle Theorems

1) Angle Sum, given  $65^\circ$ ,  $80^\circ$ .  $x = ?$

$x =$

2) Angle Sum, given  $40^\circ$ ,  $90^\circ$ .  $x = ?$

$x =$

3) Angle Sum, given  $53^\circ$ ,  $55^\circ$ .  $x = ?$

$x =$

4) Angle Sum, given  $30^\circ$ ,  $90^\circ$ .  $x = ?$

$x =$

5) Exterior Angle Theorem, remotes  $45^\circ$ ,  $75^\circ$ . Exterior  $x = ?$

$x =$

6) Exterior Angle Theorem, remote  $55^\circ$ , exterior =  $135^\circ$ .  $x = ?$

$x =$

7) Exterior Angle Theorem, remote  $46^\circ$ , exterior =  $118^\circ$ .  $x = ?$

$x =$

8) Angle Sum, given  $50^\circ$ ,  $50^\circ$ .  $x = ?$

$x =$

[← Back to Unit 4](#)